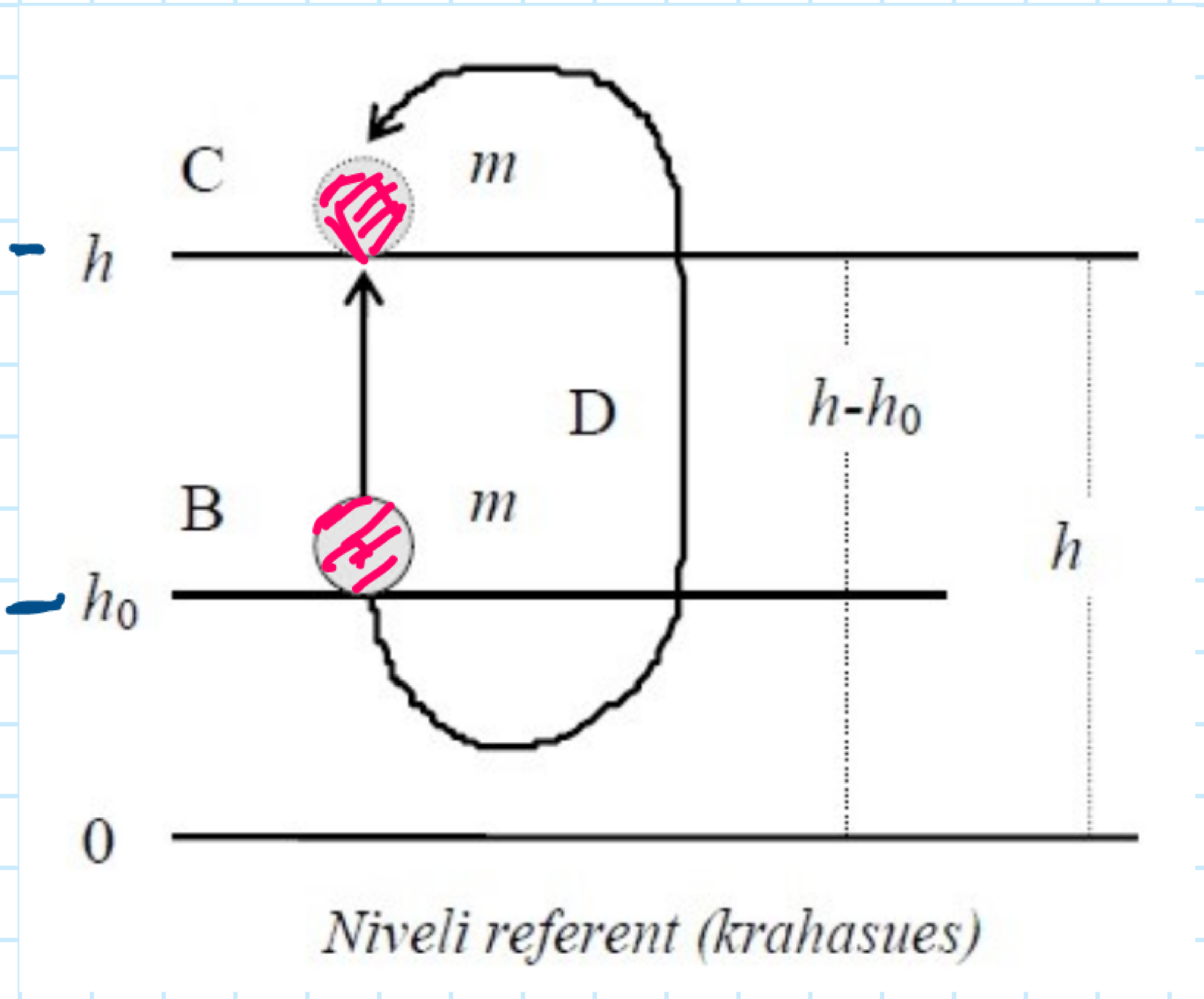
Ligjerata 6ë

Energjia potenciale e rendimit

$$A = \int_{h_0}^{h} F\,dh \qquad s = h$$



$$F = mg = Q$$

$$A = \int_{h_0}^{h} mg\,dh$$

$$A = mg\left(h\Big|_{h_0}^{h}\right)$$

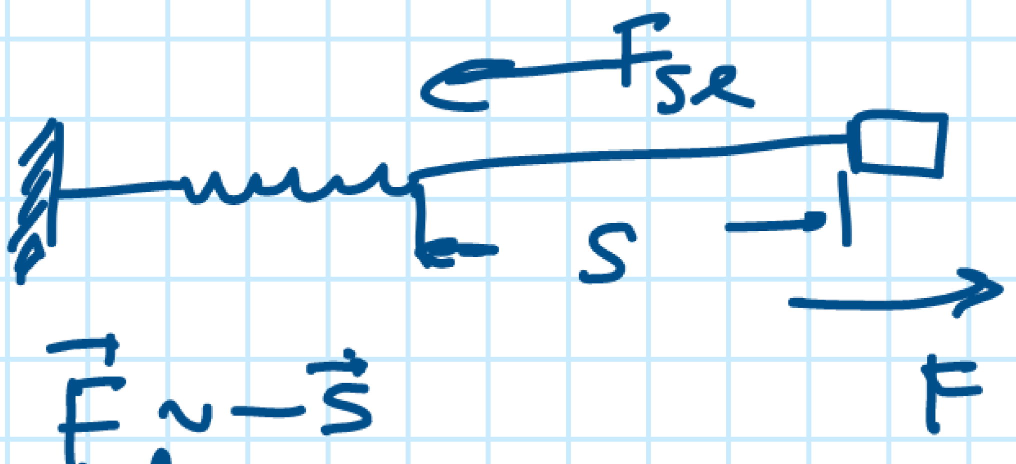
$$\oint \vec{F}\,d\vec{s} = 0$$

$$A = mg(h - h_0) = mgh - mgh_0$$

$$E_p = mgh$$

Energjia potenciale elastike

$$\Delta A_{el} = -\Delta E_{el}$$



$$\vec{F}_{el} \sim -\vec{s}$$

$$\vec{F}_{el} = -k\vec{s}$$

$$A_{el} = \int_{s_1}^{s_2} \vec{F}_{el} \cdot ds = -\int_{s_1}^{s_2} k s\,ds$$

$$A_{el} = -k\left(\frac{s^2}{2}\Big|_{s_1}^{s_2}\right) = -k\left(\frac{s_2^2}{2} - \frac{s_1^2}{2}\right)$$

$$A_{el} = -\left(\frac{k s_2^2}{2} - \frac{k s_1^2}{2}\right)$$

$$\boxed{\Delta A_{el} = -\Delta E_{el}}$$

$$\boxed{E_{el} = \frac{k s^2}{2}}$$